PUSHPAK Grand Challenge 2026

General Guidelines

- In any case, you are not allowed to have more than 5 members per team. You may add/remove members at this stage but make sure you don't exceed 5 members per team. You are required to follow competition guidelines.
- Only Design and/or software simulation is expected during stage 2, depending on the grand challenge you have qualified for.
    - Stage 2 does not require any hardware implementation. Only detailed engineering design for teams qualified for Grand Challenge 2
    - Fully functional IDS system is expected in Stage 2 for teams qualified for Grand Challenge 3 (Objective 2)
- Hardware components to set up tests to validate designs/simulations can be covered. Computational resources (Except Desktop/laptop), licenses, subscriptions, etc. can be covered as long as it can be justified for the project. External testing facilities, access to specialized equipment is permitted. Consultancy services will not be permitted.

  A storage drive (SSD or Pendrive) for simulation logs an eligible item? - Eligible

  We can allow procurement of drones or procurement of parts to build drone if you are able to justify the need

  All the above is subject to approval - you are required to submit the budget proposal on or before 18 October 2026.

- No advance payment shall be made.
- Budget proposal approval is one process - which might take 1-2 weeks
- Reimbursement is a different process once budget approved and purchase done at your end - this might take 30-45 days (Requires all invoice along with payment proof, submission link shall be shared along with budget approval).
- Reimbursement may be allowed twice during the competition and if anything is still pending that shall be considered after 02 December 2026
- Be assured, you will get fully reimbursed for all the approved ones, if expenses are made on or before 02 December 2026.
- Support approved for stage 2 shall be utilized during stage 2 only. Unused support cannot be carried forward to the next stage.
- You need approval for upto ₹1,00,000 support that shall be provided by us. whatever amount that you put in yourself need no approval from our end. and if you want to put it in the common budget proposal file then you may and in the comments please specify that the cost shall be borne by the team.
- If any changes are made after the budget is approved, Team Leader is required to seek approval over email: pushpak.gc2026@gmail.com along with the justification.
- The faculties involved in organizing the competition cannot participate as guides for any teams until the stage 3 (Grand Finale at IIT Bombay) as that would lead to conflict of interest. However, the team is free to approach other faculties.
- Stage 3 is at IIT Bombay which includes demo/presentation + viva during 16-18 December 2026 (techfest) at IIT Bombay. Build & Test phase shall be after this stage 3 (Only a few selected teams shall enter Build & Test Phase)
- In case a team leader is not possessing a PAN card then nominate one of the team members and share his/her details in the form - pls make sure all the

information in the form belongs to your nominated teammate. Also share your consent over email.

- Disbursement of support fund is purely subject to submission of valid invoices and payment proofs

Challenge Specific Question:

Grand Challenge 1

1. What machine will the re-run use (CPU cores, RAM, OS), and may the simulation run slower than real time? We will verify our submission on equivalent hardware before sending it.

I9/Ryzen 9xxxx, 16cores, 128gb ram, ubuntu. We will provide a docker and you can add you simulation into that docker.

2. Should changed parameters be accepted as scenario input before a run, or must the swarm also adapt to changes made mid-mission? This decides the interface we expose.

Some can be scenario input while others can be dynamics. Failure of a vehicle is dynamic, while area of search, sensor range/parameters are prefixed.

3. Likewise for disturbances: will you inject them with your own tools, or supply them in the scenario for our system to apply? When will the benchmark scenarios and log format be available?

We will send the docker all the log format before 20th oct. Disturbances are dynamic and you need to add those interfaces as a panel/box, whatever graphical way you want to show.

4. Some combinations are physically infeasible for a small fleet: at 100 m range over a 1 km area, four UAVs cannot form a continuous chain. In such a case is fleet size ours to set per scenario, or should the system report that the mission cannot be met?

If the input is constrained, you can flag that the mission is not possible.

5. The metrics list communication downtime and recovery time. Should we read "connected at all times" as the nominal requirement, with the outage after a failure scored through those metrics?

yes.

6. With 20 m separation at all times and UAVs returning to recharge: what are the dimensions of the operational centre, and is the transit between the centre and the operational area outside the geofence?

The operational area is 10mx 10m.

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Grand Challenge 2

Is maximum thrust the continuous limit set by the motor, electrical and structural ratings, or a short-burst peak? - Continuous thrust capability
And must 10 N and the 2.5 ratio hold at the same operating point? - Yes.

Besides the battery and vehicle mount, are the motor, speed controller, pitch servos and wiring all counted in module weight? Yes
Is mass taken from CAD with material densities plus a stated allowance for fasteners and adhesive? Yes.

Will you re-run our CFD and FEA cases or review the submitted setup and results? If re-running, with which solver and version? Only review.

Is 2D CFD with a stated 3D correction, validated against published cyclorotor experiments, an acceptable basis for the thrust claim, or is full 3D expected? Both acceptable, with a sound engineering approach.

Beyond thrust and thrust-to-weight, which quantities are assessed: vectoring range, power at 10 N, structural margin? Power loading (T/P), vectoring range, structural margin, well supported by verifiable calculations and simulations.

Does the ParaView-compatible requirement apply to simulation results only, with CAD submitted as STEP? Yes And when will the Stage 2 report template arrive? The template is already provided.

Grand Challenge 3

Objective 1

It is expected that the solution demonstrates extraction of forensics evidences, telemetry data, on board sensor data and firmware.

Objective 2

Fully functional IDS system is expected in Stage 2